

ANGCHEN XIE

Robotics Institute, Carnegie Mellon University

xieangchen808@gmail.com | angchenxie.github.io | [Google Scholar](#) | [GitHub](#) | [X](#)

EDUCATION

Carnegie Mellon University

M.S. in Robotics (MSR)

Advised by [Guanya Shi](#) and [Max Simchowitz](#)

- GPA: 4.17/4.00
- Core courses: Introduction to Robot Learning; Optimal Control and Reinforcement Learning; Computer Vision

Pittsburgh, PA

Aug. 2025 – May 2027 (expected)

Shanghai Jiao Tong University

B.Eng. in Automation

- GPA: 4.01/4.30 | Rank: 1/87

Shanghai, China

Sept. 2021 – June 2025

RESEARCH INTERESTS

I work toward general-purpose robots that combine broad learned priors with grounded models of themselves and the physical world, enabling reliable and versatile behavior across tasks, embodiments, and environments. I am drawn to methods that have to survive contact with real hardware rather than only a simulator.

PUBLICATIONS

* Equal contribution.

- [1] **[Under review, CoRL]** Angchen Xie*, Nikhil Sobanbabu*, Ishayu Shikhare, Alan Wang, Max Simchowitz, Guanya Shi. “FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control”. [arXiv](#) • [Website](#) • [Thread](#)
- [2] **[RSS 2026]** Rishabh Madan, Angchen Xie, Samantha Saak, Andres Blanco, Dohyeok Lee, Sarah Grace Brown, Yunting Yan, Mark Zolotas, Jose Barreiros, Tapomayukh Bhattacharjee. “TACTIC: Tactile and Vision Conditioned Contact-Centric Control for Whole-Arm Manipulation”. [arXiv](#) • [Website](#)
- [3] **[RSS 2026]** Pablo Ortega-Kral*, Eliot Xing*, Arthur Buckner, Vernon Luk, Junseo Kim, Owen Kwon, Angchen Xie, Nikhil Sobanbabu, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussell, Guanya Shi, Jonathan Francis, Jean Oh. “RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning”. [arXiv](#) • [Website](#) • [Code](#)
- [4] **[CoRL 2025]** Rishabh Madan, Jiawei Lin, Mahika Goel, Angchen Xie, Xiaoyu Liang, Marcus Lee, Justin Guo, Pranav N. Thakkar, Rohan Banerjee, Jose Barreiros, Kate Tsui, Tom Silver, Tapomayukh Bhattacharjee. “PrioriTouche: Adapting to User Contact Preferences for Whole-Arm Physical Human-Robot Interaction”. [arXiv](#) • [Website](#)
- [5] **[IROS 2024]** Angchen Xie, Yeqiang Qian, Weihao Yan, Chunxiang Wang, Ming Yang. “Non-repetitive: A Promising LiDAR Scanning Pattern”. [Paper](#) • [Dataset](#)

RESEARCH EXPERIENCE

LeCAR Lab, Carnegie Mellon University

Research Intern

Advised by [Guanya Shi](#) and [Max Simchowitz](#)

- Proposed a Planner-Inverse-Dynamics architecture for humanoid whole-body control that adapts to unseen terrain, payload, and actuator dynamics by finetuning only the inverse dynamics model on about two minutes of target-domain rollouts, without rewards or demonstrations. (FADA)

Pittsburgh, PA

Aug. 2025 – Present

- Built and released open-source infrastructure for cross-embodiment robot learning, validated on vision-language-action deployment across three morphologies and four hardware platforms. (*RIO*)

RL² Lab, Shanghai Jiao Tong University

Research Intern

Shanghai, China

Jan. 2025 – June 2025

Advised by [Yue Gao](#)

- Applied model-based reinforcement learning to hexapod locomotion across varied terrain and deployed the resulting policies on hardware.
- Designed a fault-tolerant controller that adapts to the failure of one or more joints, including failure modes unseen during training.
- Introduced multi-sensor fusion to sharpen the robot’s perception of terrain and of its own failure state, improving generalization on the real platform.

EmPRISE Lab, Cornell University

Research Intern

Ithaca, NY

July 2024 – Dec. 2024

Advised by [Tapomayukh Bhattacharjee](#)

- Developed a contact-centric predictive model for whole-arm manipulation that couples learned latent dynamics with analytical kinematics through contact Jacobians, and drove it with a contact-aware sampling-based MPC planner. (*TACTIC*)
- Contributed to a learning-to-rank framework that resolves conflicting user force preferences across simultaneous contacts in physical human–robot interaction. (*PrioriTouch*)

CyberC3 Intelligent Vehicle Lab, Shanghai Jiao Tong University

Research Intern

Shanghai, China

Nov. 2022 – June 2024

Advised by [Yeqiang Qian](#) and [Ming Yang](#)

- Built *Repetitive-or-not*, a CARLA dataset captured simultaneously by repetitive and non-repetitive LiDARs, isolating scanning pattern as the single variable.
- Showed that non-repetitive scanning is the more promising pattern on the strength of its object scanning, raising the maximum detection count on small traffic participants by 59.6%, and quantified the domain gap between the two point-cloud types. (*IROS 2024*)

SERVICE

Reviewer, Conference on Robot Learning (CoRL)	2026
Reviewer, Robotics: Science and Systems (RSS)	2026
Reviewer, IEEE Robotics and Automation Letters (RA-L)	2025, 2026

TALKS

<i>Non-repetitive: A Promising LiDAR Scanning Pattern</i> IROS 2024, Abu Dhabi, UAE	2024
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HONORS AND AWARDS

Outstanding Graduate Award, Shanghai Jiao Tong University	2025
China National Scholarship (top 0.2%)	2024
First Prize Academic Scholarship, Shanghai Jiao Tong University	2024
National First Prize, China Undergraduate Electronics Design Contest (top 1%)	2023
Second Prize, China Undergraduate Engineering Practice and Innovation Contest	2023
First Prize Scholarship, Shanghai Shangjun Investment Management Co.	2023
First Prize, China Undergraduate Mathematical Contest in Modeling (top 2%)	2022